

# Virtual TELEPRESENCE ROBOT Using RASPBERRY Pi

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This robot with a camera is placed in a remote location to capture the environment in visual form using Raspberry Pi (RPI). The captured visuals are displayed on the user's virtual reality (VR) headset.

An added feature allows the camera to move in the direction of the user's head movements. This gives the user a real-time experience as if he is present where the robot is located.

The robot can also be moved in any direction through an app installed in the user's smartphone.

## Virtual reality

The video captured by RPi camera can be viewed on a smartphone placed in the virtual reality (VR) headset. The project lets the user experience virtual reality through VR headset. Dual-screen mode is enabled in the smartphone for this purpose.

The smartphone reads the accelerometer and magnetometer data of the direction in which the user turns his head, say, right or left. This data is sent to the modem over Wi-Fi and to the RPi board, which, in turn, provides these values as inputs to the servo motors.

Two servo motors are used to move the camera—one for the vertical movement and the other for the horizontal movement. So when you turn your head along with VR headset to the right side, the RPi camera will also turn to the right direction.

## Circuit and working

The block diagram of virtual telepresence robot is shown in Fig. 1. RPi is the brain of the system. It receives input from the smartphone via Wi-Fi, and then sends controlling pulse to the

servo motors to move the RPi camera. The smartphone also provides

input to the Arduino Mega (Board2) for the purpose of navigation or movement of the robot. The motor driver IC and geared motors are connected at the end of the navigation circuit.

The commands to run the robot can be sent via Bluetooth or RF module from the smartphone. In this example, Bluetooth is used.

The main circuit diagram of virtual telepresence robot is shown in Fig. 2. The project has following sections: 12V charger, 8V power supply, Raspberry Pi board, Arduino

Mega 2560 board, servo motors, DC geared motors and Bluetooth module. Along with these you also require an Android smartphone with relevant app, Python and PHP codes for Raspberry Pi, and Arduino sketch for Arduino board.

**Power supply.** The first stage of the circuit is a 12V charger for charging rechargeable batteries. Four lead-acid rechargeable batteries, each of 4V, 1Ah, are arranged in series and parallel combination to provide 8V power supply. If the batteries are connected in series, the final voltage gets added up and current remains the same. If the batteries are connected in parallel, current adds up and voltage remains the same. Thus, you get 8V, 2Ah power supply by connecting two each in series and then the two sets in parallel.

Diode D1 is used to block discharging of the battery through LED1 and capacitor C1. The diode also makes the circuit stable and controls temperature while charging.

The 8V power supply is used to power the Arduino, motor driver IC and other modules. When switch S1 is open, the battery supply to servos is cut off. Three series diodes (D2 through D4) reduce the voltage to around 6V for safe operation of servo motors.

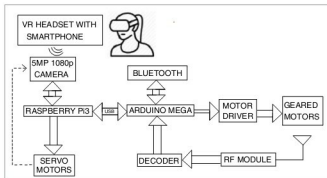


Fig. 1: Block diagram of virtual telepresence robot

## PARTS LIST

| Semiconductors:                                                      |                                                         |
|----------------------------------------------------------------------|---------------------------------------------------------|
| IC1                                                                  | - L293D motor driver                                    |
| D1-D4                                                                | - 1N4001 rectifier diode                                |
| LED1                                                                 | - 5mm LED                                               |
| Board1                                                               | - Raspberry Pi 3 with Raspbian software                 |
| Board2                                                               | - Arduino Mega 2560                                     |
| Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise): |                                                         |
| R1                                                                   | - 2.1-kilo-ohm                                          |
| Capacitors:                                                          |                                                         |
| C1                                                                   | - 0.1 $\mu$ F ceramic                                   |
| Miscellaneous:                                                       |                                                         |
| S1                                                                   | - On/off switch                                         |
| CON1                                                                 | - 2-pin connector for 8V DC jack                        |
| CON2, CON3                                                           | - 3-pin connector for servo motor                       |
| CON4                                                                 | - DC jack for 12V charger                               |
| CON5                                                                 | - 6-pin female connector for Bluetooth                  |
|                                                                      | - 7x2-pin bergstrip male connector                      |
| M1, M2                                                               | - Servo motors                                          |
| M3, M4                                                               | - 8V-12V, 50RPM, DC geared motor                        |
| BATT1-BATT4                                                          | - 4V, 1Ah lead-acid battery                             |
|                                                                      | - USB cable for connecting Board1 and Board2            |
|                                                                      | - 12V, 1A power adaptor for charging batteries          |
|                                                                      | - 5V, 1000mAh power bank for Raspberry Pi               |
|                                                                      | - Micro SD card for Raspberry Pi with Raspbian software |
|                                                                      | - RPi camera                                            |
|                                                                      | - Suitable chassis for the robot                        |
|                                                                      | - Bluetooth module (HC-05)                              |
|                                                                      | - Wheels for the robot                                  |